Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

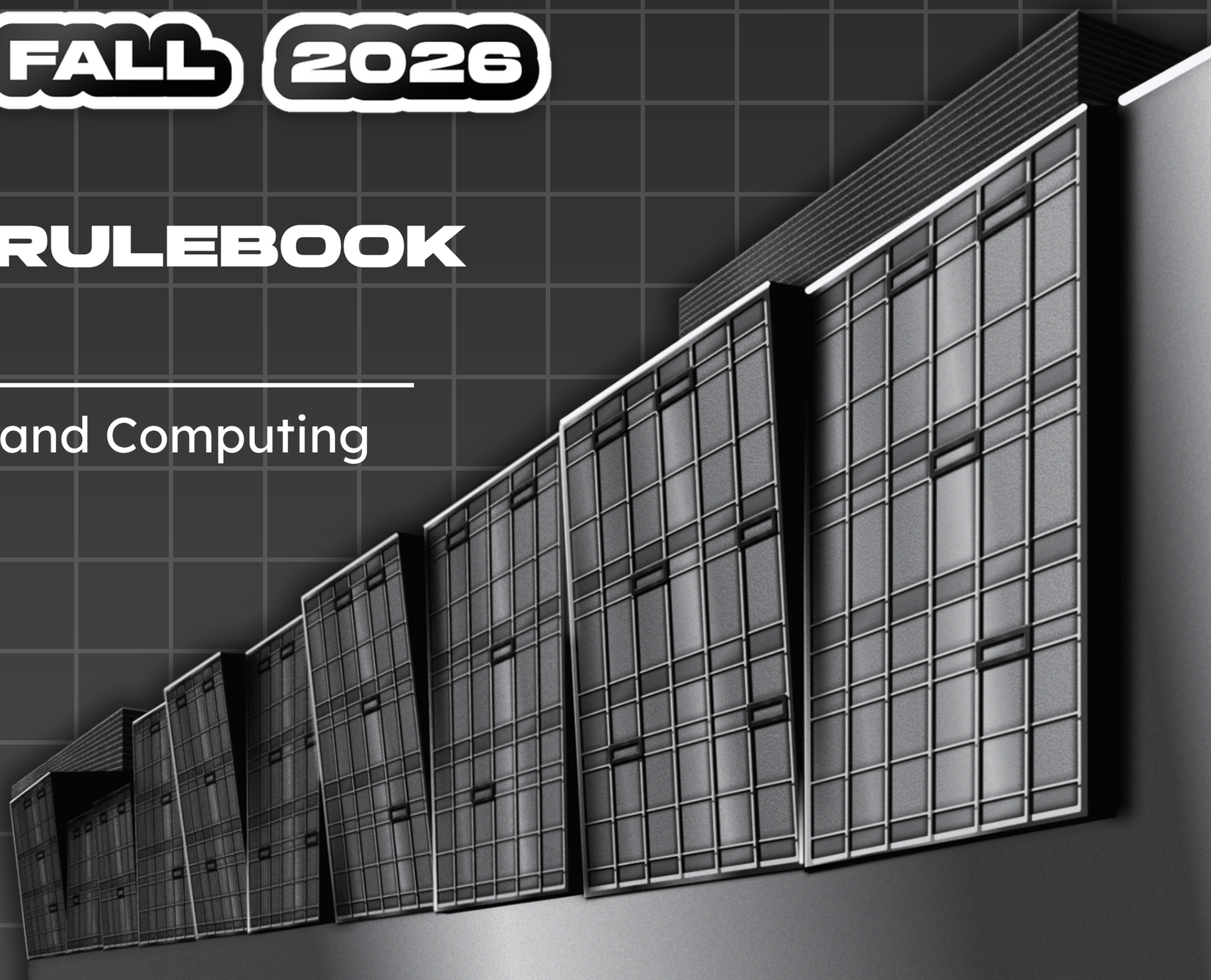
# TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

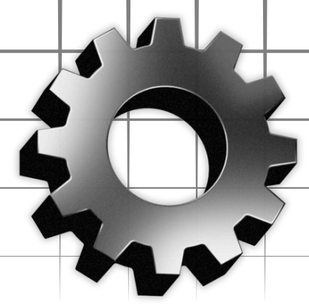
# FALL 2026

## CONSULTING RULEBOOK

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre





## 1.1 GOAL

The purpose of the Consulting Engineering competition is to challenge competitors to design a detailed solution to a large-scale engineering problem. The proposal must be made in a way that promotes the solution to the client (in the form of judges). Competitors must demonstrate resourcefulness while acting in good faith with the spirit of the competition.

## 1.2 TEAM COMPOSITION

Each Consulting Engineering team must consist of a minimum of three (3) and a maximum of four (4) registered competitors.

## 1.3 TOPIC

It is recommended that the topic incorporate more than one engineering discipline. However, while the topic should challenge competitors' technical knowledge and skills, it should also require competitors to evaluate the economic, environmental, political and social implications of their proposed solutions and address the requirements of the customer. The winning solution will not necessarily be the most technically effective solution, but the solution that has the most real-world applicability and forethought. Therefore, the topic should be one that could exist in the real world. Topics drawn from reality must be fully documented. All necessary documentation must be provided to the competitors when the problem is presented.

## 1.3 LANGUAGE

The written report and oral presentation must be delivered in English.

## 1.4 RESOURCES

### 1.4.1. Facilities Required

The Competition Organizers will provide suitable spaces for the competition briefing, team solution development, and judging presentations. Room assignments will be communicated to competitors before or during the competition.

### 1.4.2. Personnel Required

#### 1.4.2.1. Judges

A judging panel will evaluate each team's proposed solution, technical analysis, written report, presentation, and responses to questions using the official Consulting Competition Scorecard. Judges should have relevant experience in engineering, consulting, business, communication, or related fields. Judges in this category should come from a variety of backgrounds including communications, sales and technical or consulting engineering experience related to the topic.

#### 1.4.2.2. Competition Leads

Only authorized Consulting Competition Coordinators or Competition Leads may provide official clarification regarding the competition problem, rules, and deliverables. The Competition Leads

must present the design problem at the beginning of the competition and answer any questions raised by competitors. Only the Competition Leads may answer questions during the design period. The Competition Leads will also answer any questions the judges may have during the competition.

#### 1.4.2.3. Official Timekeeper

The Official Timekeeper must be responsible for enforcing time limits during the question period, solution development and presentations, as stated in the rules.

### 1.4.3. Equipment Provided

The following equipment will be provided to all competitors during the competition time period.

#### 1.4.3.1. During Design Phase

- A workspace
- A method to submit the team presentation
- Internet Connectivity
- Information relevant to the design problem, at the discretion of the Competition Coordinator

#### 1.4.3.2. During Presentation

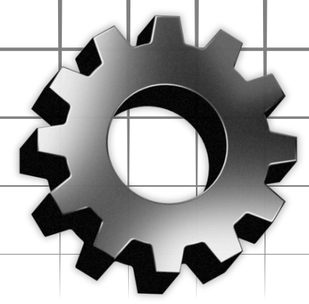
- One (1) digital projector
- Access to presentation equipment capable of displaying the team's submitted presentation file.
- Whiteboard(s) or blackboard(s)

### 1.4.4. Allowed External Resources

The competitors can bring the following equipment with them:

- Any background research conducted by the team prior to the competition
- Electronic reference materials, datasets, and files stored on personal devices or approved storage platforms.
  - Please check with the organizers to make sure the format of your electronic information will be accessible using the computers provided by MEC
- Any textbooks, course notes or other reference material
- Competitors may use their own laptops and other approved electronic devices for research, analysis, report preparation, and presentation development.

Competitors may use publicly available technical references, engineering software, spreadsheets, and presentation tools, provided all external sources are properly acknowledged. The use of generative AI tools must comply with the official MEC competition AI policy.



Note: Since the use of the internet and other external resources is permitted in this competition, all information used by competitors must be referenced very carefully. Competitors are not permitted to submit work completed by anyone other than the members of their team. If there is any evidence that competitors are submitting plagiarized work, the entire team will be eliminated from the competition. Volunteers will monitor each team during the design process to deter teams from cheating, however, competitors are expected to act in good faith with the spirit of the competition.

## 1.5 PROCEDURE/TIMELINE

### 1.5.1. Pre-Competition

Prior to the competition, the official competition rules and relevant information will be communicated to registered competitors through authorized MEC communication channels.

### 1.5.2. Competition

#### 1.5.2.1. Presentation of Problem

The problem must be presented to all competitors and judges at the beginning of the competition. The Coordinator(s) must provide detailed explanations of what is expected from the competitors, both orally and in writing.

#### 1.5.2.2. Question Period

Competitors then have fifteen (15) minutes to ask the Coordinators any questions. Only the time used to ask the questions (not the answers) should be counted in the fifteen (15) minutes. Time used by judges to pose questions will not be counted.

#### 1.5.2.3. Solution Development

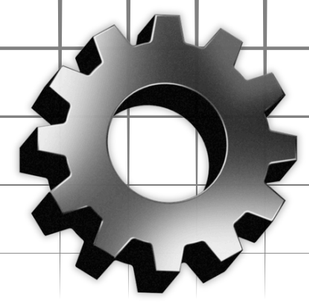
Teams will be given eight (8) hours to develop their proposed solutions, complete the required deliverables, and prepare their presentations. All deliverables must be submitted electronically through the official submission method before the end of the designated development period. Competitors may finish before the allotted time expires.

#### 1.5.2.4. Rest Period

Competitors must be allowed a minimum of one (1) hour to rest before the presentation phase starts.

#### 1.5.2.5. Solution Presentations

Competitors will have a maximum of ten (10) minutes to present their proposed solutions. All registered team members are expected to attend and participate in the presentation. Any absence-related penalties will follow the official competition penalty matrix. Judges will then have a maximum of five (5) minutes to ask questions. Judges can ask a question at any time during the presentation. The clock must be stopped during these interruptions. In order to ensure that all



competitors cease to work on the case solutions once the design time has ended, the competitors cannot include any material in their oral presentation which is not included within their submitted written reports or presentation materials.

### 1.5.3. Timekeeping

The following rules will be adhered to with respect to timekeeping:

#### 1.5.3.1. During Design Phase

- Time is started when the teams leave the competition briefing
- The remaining time must be announced three (3) hours, one (1) hour, thirty (30) minutes, and ten (10) minutes before the end of the allotted time

#### 1.5.3.2. During the Presentation

- If a judge interrupts the prepared presentation with a question, the presentation timer will be paused during the question and response. The separate five (5) minute Q&A period will be timed continuously unless the Competition Coordinators authorize a pause.
- The remaining presentation time must be indicated to competitors five (5) minutes and one (1) minute before the end of the allotted presentation period.
- A countdown presentation that is visible to both the judges and presenters must be given during the last thirty (30) seconds of the presentation.

### 1.5.4. Presentation Order

- Presentation order shall be determined randomly
- Presentation order shall be announced thirty (30) minutes before the presentations commence. All teams are required to be present at this announcement
- Teams are not allowed to switch places in the presentation order
- Teams may not be in the audience for presentations of their competition until after they have completed their presentation. Any attempts at entering the presentation room before their allotted time will result in a warning for a first offense, and a ten (10) point penalty for any further offenses

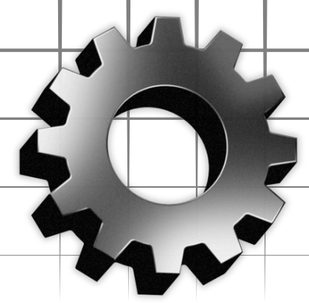
### 1.5.5. Deliverables

#### 1.5.5.1. For Competitors

Prior to the competition, registered competitors will receive the official competition rules, abstract, and relevant instructions through authorized MEC communication channels.

#### 1.5.5.2. From Competitors

Each team must electronically submit one written consulting report before the end of the designated development period. The report must not exceed ten (10) pages of main content,



excluding the title page and references. Any appendices must contain only brief supporting information and must not be used to bypass the page limit. The report must follow the requirements outlined in the official competition problem package. The report will be made available to the judges before each team's presentation. The report requirements will be outlined in the problem design package.

Each team must electronically submit one presentation slide deck before the end of the designated development period. The report and presentation must be submitted in PDF format through the official submission method communicated by the Competition Coordinators. The requirements of the presentation will be outlined in the problem design package.

#### 1.5.6. Response to Questions

- Only the Coordinators may answer questions during the problem, presentation and design phases. Volunteers and judges may not answer questions regarding rules and procedures at any time
- During the presentation of the problem, the Competition Lead shall answer orally and write down the answers provided
- During the design phase, Competition Coordinators may clarify competition rules, deliverable requirements, and ambiguities in the provided case materials. Coordinators must not provide solution strategies or technical assistance that would give any team an unfair advantage. No answers shall be provided in response to questions about the problem that might lead to the development of a new approach or which might invalidate a solution
- During the design phase, answers to questions shall be provided in writing to all teams at the same time. Fifteen (15) minutes may be allowed to elapse for a significant number of questions accumulated before responses are provided. A copy of the responses must be provided to the judges prior to the presentations

## 1.6. ASSESSMENT AND JUDGING

### 1.6.1. Judging Matrix

Competitors will be evaluated using the official Consulting Engineering Competition Scorecard provided by the Competition Organizers. The scorecard will outline the judging categories, evaluation criteria, and maximum points available for each criterion. Teams will be scored out of one hundred

(100) points before any applicable deductions. The official scorecard will serve as the primary reference for judging and determining competition results.

| Category | Judging Criteria | Points |
|---|---|---|
| Solution | A. Problem Understanding & Analysis: Demonstrates a clear understanding of the engineering problem, client requirements, stakeholder needs, and underlying challenges. | /15 |
| | B. Engineering Feasibility & Implementation: Evaluates the practicality, technical feasibility, financial considerations, and implementation strategy of the proposed solution. | /15 |
| | C. Equity & Accessibility: Considers accessibility, inclusivity, and the needs of different stakeholders or affected communities. | /10 |
| | D. Trade-offs & Risk Analysis: Identifies and justifies engineering trade-offs, potential risks, and appropriate mitigation strategies. | /10 |
| | E. Long-Term Sustainability & Innovation: Evaluates long-term sustainability, maintainability, scalability, and the originality of the proposed solution. | /10 |
| Solution Subtotal | | /60 |
| Report | A. Organization & Professionalism: Evaluates the report's structure, clarity, formatting, and professional presentation. | /5 |
| | B. Technical Evidence & Supporting Analysis: Assesses the quality of calculations, supporting evidence, technical analysis, and appropriate use of references. | /5 |
| | C. Recommendation & Design Justification: Evaluates how clearly the final recommendation is supported by engineering reasoning and evidence. | /5 |
| Report Subtotal | | /15 |

| | | |
|---|---|---|
| Presentation | A. Delivery & Professional Communication: Evaluates speaking clarity, confidence, professionalism, and the ability to communicate effectively with the judging panel. | /10 |
| | B. Organization & Visual Communication: Assesses presentation structure, logical flow, visual aids, and clarity of information. | /5 |
| | C. Technical Understanding & Q&A: Evaluates technical knowledge, justification of decisions, and responses to judges' questions. | /5 |
| | D. Team Participation & Time Management: Assesses meaningful participation by team members and effective use of the allotted presentation time. | /5 |
| Presentation Subtotal | | /25 |
| TOTAL | | /100 |

### 1.6.2. Penalties

| | | |
|---|---|---|
| **Point Penalties** | Confirmed plagiarism or deliberate misrepresentation of another person's work as original | Elimination |
| | Insufficient citations or failure to properly acknowledge external sources | -50 |
| | Documents received after the official submission deadline without an approved extension | -50 |
| | Absent team member without an approved exception | -25 |
| | Entering the presentation room before the assigned time without authorization | First offence: Warning; each subsequent offence: -10 points |